Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Locate the centre of mass of a uniform semicircular ring (or wire) of radius R and linear mass density λ.

(A) $\frac{3R}{\pi}$

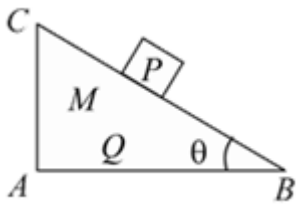
(B) $\frac{4R}{\pi}$

(C) $\frac{6R}{\pi}$

(D) $\frac{2R}{\pi}$

Q.2

A block Q of mass M is placed on a horizontal frictionless surface AB and a body P of mass m is released on its frictionless slope. As P slides by a length L on this slope of inclination θ, the block Q would slide by a distance



(A) $\frac{m}{M}L\cos\theta$

(B) $\frac{m}{M+m}L$

(C) $\frac{M+m}{mL\cos\theta}$

(D) $\frac{mL\cos\theta}{m+M}$

Q.3

A bead can slide on a smooth straight wire and a block of mass m is attached to the bead by a light string of length L. The block is held in contact with the wire with the string taut and is then let fall. The bead has mass $2\ \mathrm{m}$. When the string makes an angle θ with the wire the bead will have slipped a distance:

(A) $L(1-\cos\theta)$

(B) $\frac{L}{2}(1-\cos\theta)$

(C) $\frac{L}{3}(1-\cos\theta)$

(D) $\frac{L}{6}(1-\cos\theta)$

Q.4

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A pulley fixed to the ceiling carries a string with blocks of mass m and $3\ \mathrm{m}$ attached to its ends. The masses of string and pulley are negligible. When the system is released, its centre of mass moves with what acceleration ?

(A) 0

(B) $-\mathrm{g}/4$

(C) $\mathrm{g}/2$

(D) $-\mathrm{g}/2$

Q.5

Two spheres A and B of masses m and $2m$ and radii $2R$ and R respectively are placed in contact as shown. The COM of the system lies



(A) inside A

(B) inside B

(C) at the point of contact

(D) None of these

Q.6

There are some passengers inside a stationary railway compartment. The centre of mass of the compartment itself (without the passengers) is C_1, while the centre of mass of the 'compartment plus passengers' system is C_2. If the passengers move about inside the compartment then

(A) both C_1 and C_2 will move with respect to the ground

(B) neither C_1 nor C_2 will be stationary with respect to the ground

(C) C_1 will move but C_2 will be stationary with respect to the ground

(D) C_2 will move but C_1 will be stationary with respect to the ground

Q.7

Three masses are placed on the x -axis : $300\ \mathrm{g}$ at origin, $500\ \mathrm{g}$ at $x=40\ \mathrm{cm}$ and $400\ \mathrm{g}$ at $x=70\ \mathrm{cm}$. The distance of the centre of mass from the origin is

(A) $40\ \mathrm{cm}$

(B) $45\ \mathrm{cm}$

(C) $50\ \mathrm{cm}$

(D) $30\ \mathrm{cm}$

Q.8

With O as the origin of the coordinate axis, the X and Y-coordinates of the centre of mass of the system of particles shown in the figure may be given as? (Here m and $2m$ represent the masses of the particles)

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(A) $\left(-\frac{b}{2}, 0\right)$

(B) $\left(-\frac{b}{2}, h\right)$

(C) $\left(-\frac{b}{3}, h\right)$

(D) $\left(-\frac{2}{5}b, h\right)$

Q.9

Four particles of masses m_1, m_2, m_3 and m_4 are placed at the vertices $A, B\ C$ and D as respectively of a square shown. The COM of the system will lie at diagonal AC if



(A) $m_1 = m_3$

(B) $m_2 = m_4$

(C) $m_1 = m_2$

(D) $m_3 = m_4$

Q.10

The centre of mass of three bodies each of mass $1\ \text{kg}$ located at the points $(0, 0), (3, 0)$ and $(0, 4)$ in the XY plane is

(A) $\left(\frac{4}{3}, 1\right)$

(B) $\left(\frac{1}{3}, \frac{2}{3}\right)$

(C) $\left(\frac{1}{2}, \frac{1}{2}\right)$

(D) $\left(1, \frac{4}{3}\right)$

Q.11

$(\text{n} - 1)$ equal point masses each of mass m are placed at the vertices of a regular n-polygon. The vacant vertex has a position vector a with respect to the centre of the polygon. The position vector of centre of mass is

(A) $-\frac{1}{(n-1)}a$

(B) $\frac{a}{(n+1)}$

(C) $\frac{a}{n}$

(D) $\frac{n}{a+1}$

Q.12

If the linear density (mass per unit length) of a rod of length $3\ \text{m}$ is proportional to x, where x is the distance from one end of the rod, the distance of the centre of gravity of the rod from this end is

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(A) $2.5\ \mathrm{m}$

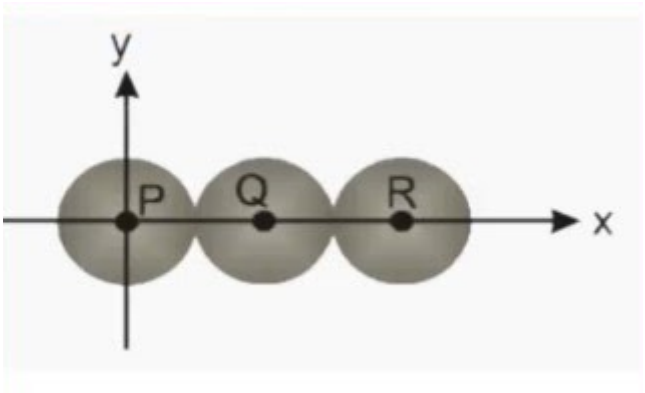
(B) $1\ \mathrm{m}$

(C) $1.5\ \mathrm{m}$

(D) $2\ \mathrm{m}$

Q.13

Three identical spheres, each of mass $1\ \mathrm{kg}$ are kept as shown in figure, touching each other, with their centres on a straight line. If their centres are marked $\mathrm{P}, \mathrm{Q}, \mathrm{R}$ respectively, the distance of centre of mass of the system from P is



(A) $\frac{PQ+PR+QR}{3}$

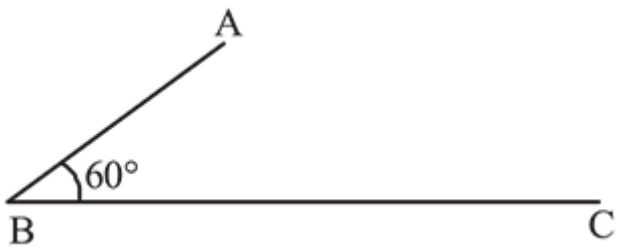
(B) $\frac{PQ+PR}{3}$

(C) $\frac{PQ+QR}{3}$

(D) $\frac{PR+QR}{3}$

Q.14

In the figure shown ABC is a uniform wire. If centre of mass of wire lies vertically below point A, then $\frac{\mathrm{BC}}{\mathrm{AB}}$ is close to:



(A) 1.85

(B) 1.5

(C) 1.37

(D) 3

Q.15

A circular disc of radius R is removed from a bigger circular disc of radius $2\mathrm{R}$ such that the circumferences of the discs coincide. The centre of mass of the new disc is α/R form the centre of the bigger disc. The value of α is

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(A) $1/4$

(B) $1/3$

(C) $1/2$

(D) $1/6$

Q.16

Three identical rods are hinged at point A as shown. The angle made by rod AB with vertical is



(A) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(B) $\tan^{-1}\left(\frac{3}{4}\right)$

(C) $\tan^{-1}(1)$

(D) $\tan^{-1}\left(\frac{4}{3}\right)$

Q.17

A spool is pulled horizontally by two equal
the following statements are correct?



(A) The centre of mass moves towards left.

(B) The centre of mass moves towards right

(C) The centre of mass remains stationary

(D) The net force about the centre of mass of the spool is zero

Q.18

The mass per unit length of a non-uniform rod of length L varies as $m = \lambda x$ where λ is constant. The centre of mass of the rod will be at :

(A) $\frac{2}{3}L$

(B) $\frac{3}{2}L$

(C) $\frac{1}{2}L$

(D) $\frac{4}{3}L$

Q.19

A right circular cone of base diameter $3\ \mathrm{cm}$ and height $6\ \mathrm{cm}$ is cut from a solid cylinder of diameter $5\ \mathrm{cm}$ and height $12\ \mathrm{cm}$. Find the position of CG of rest of the body.

(A) 2.1 cm

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(B) 6.3 cm

(C) 8.2 cm

(D) 5.3 cm

Q.20

A solid sphere of radius R is placed on a smooth horizontal surface. A horizontal force F is applied at height h from the lowest point. For the maximum acceleration of the centre of mass,

(A) h=R

(B) h=2 R(c)

(C) h=0

(D) The acceleration will be same whatever h may be

Answers & Solutions

Q.1 Answer:

$\frac{2R}{\pi}$

Solution:

Let us take the centre of the ring at origin O. Consider a small element of arc length dl of the ring. Let θ be the angle which the radius vector of the element makes with the x-axis as shown in Fig.
Let $d\theta$ be the angle subtended by the element at the centre. Then $dl = Rd\theta$. Mass of the element is $dm = \lambda dl = \lambda R d\theta$
The x and y components of radius vector R are $x = R\cos\theta$ and $y = R\sin\theta$. Then

$$x_{\text{CM}} = \frac{\int x dm}{\int dm} = \frac{\lambda R^2 \int_0^\pi (\cos\theta) d\theta}{\lambda R \int_0^\pi d\theta}$$

$$= \frac{\lambda R^2}{\lambda R \pi} |\sin\theta|_0^\pi$$

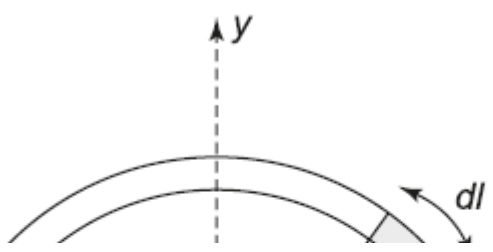
$$= \frac{R}{\pi}(\sin\pi - \sin 0) = 0$$

$$y_{\text{CM}} = \frac{\int y dm}{\int dm} = \frac{\lambda R^2 \int_0^\pi (\sin\theta) d\theta}{\lambda R \int_0^\pi d\theta}$$

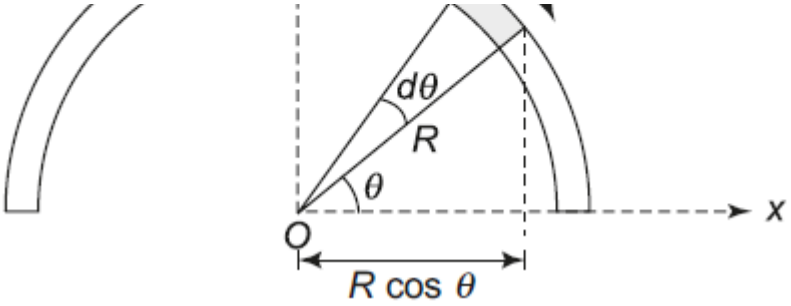
$$= \frac{\lambda R^2}{\lambda R \pi} |-\cos\theta|_0^\pi$$

$$= -\frac{R}{\pi}(\cos\pi - \cos 0) = \frac{2R}{\pi}$$

Thus, the centre of mass is at a distance of $\frac{2R}{\pi}$ from origin O on the y-axis. By symmetry, the x-coordinate of centre of mass is $x = 0$ (i.e. at O).



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Q.2 Answer:

$\frac{mL\cos\theta}{m+M}$

Solution:

Here, the x co-ordinate of centre of mass of the system remains unchanged when the mass m moved a distance $L\cos\theta$, let the mass $(m+M)$ moves a distance x in the backward direction.

$\therefore (M+m)x - mL\cos\theta = 0 \quad \therefore \quad x = \frac{mL\cos\theta}{m+M}$

Q.3 Answer:

$\frac{L}{3}(1-\cos\theta)$

Solution:

$s = \frac{m(L(1-\cos\theta))}{3m} = \frac{L}{3}(1-\cos\theta)$

Q.4 Answer:

$-\mathrm{g}/4$

Solution:

The acceleration of the whole system can be found by force balance i.e,

$(3\ \mathrm{m}+\mathrm{m})\mathrm{a}_{\mathrm{sys}} = 3\mathrm{mg} - \mathrm{mg}$

$\therefore\ \mathrm{a}_{\mathrm{ays}} = \frac{g}{2}$

$\text{so, } \mathrm{a}_{\mathrm{cm}} = \frac{\mathrm{m}_1\mathrm{a}+\mathrm{m}_2\mathrm{a}}{\mathrm{m}_1+\mathrm{m}_2} = \frac{(\mathrm{m}+(-3\ \mathrm{m}))\frac{\mathrm{g}}{2}}{\mathrm{m}+3\ \mathrm{m}}$

$\therefore\ \mathrm{a}_{\mathrm{mm}} = \frac{-\mathrm{g}}{4}$

Q.5 Answer:

at the point of contact

Solution:

Let $\mathrm{O}(0,0)$ be the centre of sphere A then, $\chi_{\mathrm{cm}} = \frac{\mathrm{m}\times 0+2\ \mathrm{m}\times 3\mathrm{R}}{\mathrm{m}+2\ \mathrm{m}} = 2\mathrm{R}$

$=$ at the point of contact

Q.6 Answer:

C_1 will move but C_2 will be stationary with respect to the ground

Solution:

When net $F_{\text{atermal}} = 0$, then the centre of mass of the system remains at rest. Thus if the passenger

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move about inside the compartment which donot require any external force, so the centre of mass of the "passenger + compartment" system must remain at rest and hence C_2 will be fixed w.r.t ground.
Also due to the movement of the passenger, the position of centre of mass of the passengers only will change, thus C_1 will have to move in such a way that C_2 may remain fixed w.r.t ground.

Q.7 Answer:

40 cm

Solution:

$$X_{\text{cm}} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$
$$X_{\text{cm}} = \frac{300 \times (0) + 500(40) + 400 \times 70}{300 + 500 + 400}$$
$$X_{\text{cm}} = \frac{500 \times 40 + 400 \times 70}{1200} = 40 \text{ cm}$$

Q.8 Answer:

$\left(-\frac{b}{3}, h\right)$

Solution:

$\bar{x} = \frac{m(-2b)+2m(-b)+m\times 0+2m(b)}{m+2m+m+2m} = \frac{-b}{3}$
and $\bar{y} = +h$**.**

Q.9 Answer:

$m_2 = m_4$

Solution:

Possible when $m_2 = m_4$

Q.10 Answer:

$\left(1, \frac{4}{3}\right)$

Solution:

$$X_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$
$$= \frac{1 \times 0 + 1 \times 3 + 1 \times 0}{1 + 1 + 1} = 1$$
$$Y_{CM} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$
$$= \frac{1 \times 0 + 1 \times 0 + 1 \times 4}{1 + 1 + 1} = \frac{4}{3}$$

Therefore the coordinates of centre of mass are $\left(1, \frac{4}{3}\right)$ **.**

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$$X_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$= \frac{1 \times 0 + 1 \times 3 + 1 \times 0}{1 + 1 + 1} = 1$$

(d)

$$Y_{CM} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

$$= \frac{1 \times 0 + 1 \times 0 + 1 \times 4}{1 + 1 + 1} = \frac{4}{3}$$

Therefore the coordinates of centre of mass are $\left(1, \frac{4}{3}\right)$.

Q.11 Answer:

$-\frac{1}{(n-1)}a$

Solution:

Let the C.M. be 'b'. Then, $\frac{(n-1)mb+ma}{mn} = 0 \Rightarrow b = -\frac{1}{n-1}a$

Q.12 Answer:

2 m

Solution:

Consider an element of length dx **at a distance** x **from end** A**.**

Here, mass per unit length λ **of rod**

$\lambda \propto x \Rightarrow \lambda = kx$

$\therefore dm = \lambda dx = kxdx$

Position of centre of gravity of rod from end A**.** $x_{CG} = \frac{\int_0^L xdm}{\int_0^L dm}$

$\therefore x_{CG} = \frac{\int_0^3 x(kxdx)}{\int_0^3 kxdx} = \frac{\left[\frac{x^3}{3}\right]_0^3}{\left[\frac{x^2}{2}\right]_0^3} = \frac{\frac{(3)^3}{3}}{\frac{(3)^3}{2}} = 2m$

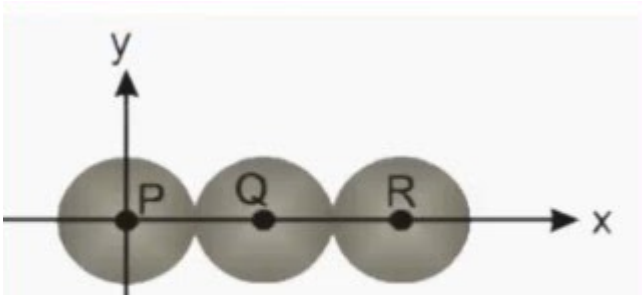
Q.13 Answer:

$\frac{PQ+PR}{3}$

Solution:

$x_{cm} = \frac{1\times0+1\times PQ+1\times PR}{1+1+1} = \frac{PQ+PR}{3}$

and $y_{cm} = 0$



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Q.14 Answer:

1.37

Solution:

Centre of mass $x_{cm} = \frac{x}{2}\frac{(\rho x)\left(\frac{x}{2}\right)\frac{1}{2}+\rho y\left(\frac{y}{2}\right)}{\rho(x+y)} \Rightarrow \frac{1}{2}+\frac{y}{x}=\frac{y^2}{x^2}$

$\therefore \frac{BC}{AB}=\frac{y}{x}$

$=\frac{1+\sqrt{3}}{2}=1.37$

Q.15 Answer:

$1/3$

Solution:

$X_{CM}=\frac{\sigma\pi(2R)^2\times 0-\sigma\left(\pi R^2\right)R}{\sigma\pi(2R)^2-\sigma(\pi R)^2}$

$\frac{R}{3} \Rightarrow \alpha = 1/3$

Q.16 Answer:

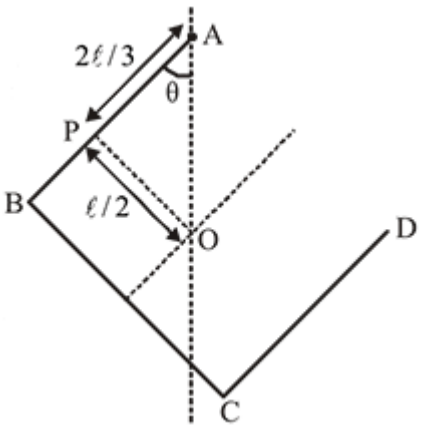
$\tan^{-1}\left(\frac{3}{4}\right)$

Solution:

Vertical line from hinge A must pass through C.M. of rod system.

$\tan\theta=\frac{OP}{AP}=\frac{\ell/2}{2\ell/3}$

$\tan\theta=\frac{3}{4}\Rightarrow\theta=\tan^{-1}\left(\frac{3}{4}\right)$



Q.17 Answer:

The centre of mass moves towards right

Solution:

The surface is rough and the couple is acting on the spool, the orientation of the spool will make the friction act on the spool towards right. Thus, the spool will move in the rightward direction.

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Q.18 Answer:

$\frac{2}{3}L$

Solution:

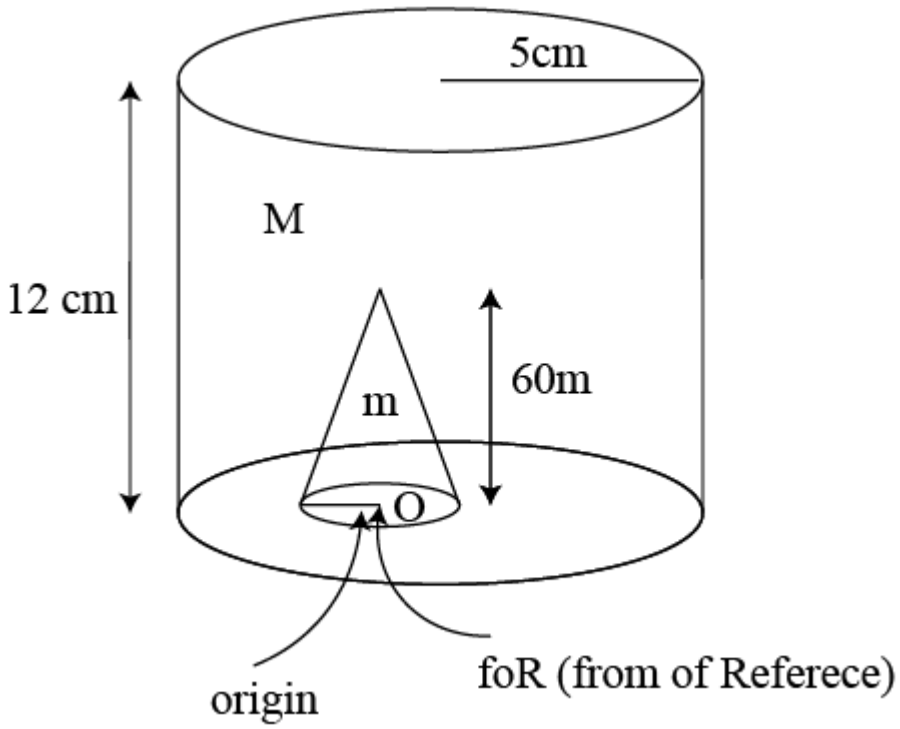
$$X_{\text{cm}} = \int_0^L \frac{(dm)x}{\int_0^L (dm)} = \frac{\int_0^L (\lambda x dx)x}{\int_0^L (\lambda x dx)} = \frac{2L}{3}$$

Q.19 Answer:

6.3 cm

Solution:

$$M = PV_{01} M = P\pi r_0^2 l$$
$$= P\pi 300$$
$$m = PV_{01m} = P \cdot \pi x^{2n}$$
$$= 1\pi/8$$
$$X_{\underset{(PROT)}{ROT}} = \frac{MX_1 - mX_2}{M-m}$$
$$x_{\text{com}} = 6 \times 1/300 - 1\pi 18 \times 2 \quad x_2 = \frac{h}{3} = \frac{6}{3} = 2$$
$$300hx - 18f5$$
$$= \frac{300\times 6 - 18\times 2}{300-18} = \underline{\underline{6.3}}$$



Q.20 Answer:

The acceleration will be same whatever h may be

Solution:

As friction is absent at the point of contact, Accelerati on $= \frac{\text{Force}}{\text{Mass}}$ **It is independent of** h

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